

Session 7: Exercises (participant version)

Foundations of Clinical Research · Session 7 (Week 7, Saturday): Protect & write (ethics, integrity, Methods and Results)

Part	Exercise	When	Time
A	Integrity scenarios: green, amber or red?	0:55	5 min (3 in class, rest at home)
B	Proposal builder 7: ethics, registration, feasibility, publication plan	1:00	15 min, finish at home
C	[L2] Manuscript workshop 1: Methods + Results of your FLUID-ICU paper	2:05	50 min

Part A: Integrity scenarios: green, amber or red? (5 minutes, 0:55; three in class, the rest at home)

For each scenario, decide as a group and hold up a colour:

- **GREEN:** acceptable as described.
- **AMBER:** acceptable **only if** something is fixed or disclosed (say what).
- **RED:** research misconduct or a serious breach. Do not do it.

Write one sentence explaining your colour.

#	Scenario	Colour	Why?
1	Three patients' day-3 fluid charts are lost. A resident "remembers roughly" what their balances were and enters values so the dataset is complete.		
2	After seeing the results, the team removes two patients with "strange" fluid values because they weaken the association. The paper does not mention it.		
3	The head of department did not take part in the study but asks to be last author "because it is my department".		

#	Scenario	Colour	Why?
4	(a) A nurse uses an AI chatbot to improve the English of the Discussion she wrote, checks every sentence, and states this in the Acknowledgements. (b) A colleague uploads the study spreadsheet (names and hospital numbers included) to a free public chatbot and asks it to "run the statistics".	(a)	
		(b)	
5	A cohort study of ventilated patients is published as three papers (mortality, length of stay and ventilator days) in three journals, none mentioning the others.		
6	Two paragraphs of a published review are copied word for word into your Introduction. The review is cited at the end of the paragraph, but there are no quotation marks.		

Part B: Proposal builder 7: ethics, registration, feasibility and publication plan (15 minutes, 1:00)

Complete as a group in 15 minutes (B1–B3 about 8 minutes, B4 about 7 minutes); finish at home during the week. Slides 5 (ethics & feasibility) and 6 (publication plan) of next Saturday's presentation come from this sheet.

B1. Ethics and consent

Question	Your answer
Which ethics committee will review the study? What type of review do you expect (exempt / expedited / full)? The committee decides.	
Who are the participants, and can they decide for themselves when they become eligible?	
Who gives consent (patient / legal representative / deferred consent / waiver requested)? Use the ICU consent flowchart.	
Who takes consent, when and where?	
What are the risks and burdens for participants, and how are they minimised?	

Question	Your answer
Does the study need registration (trials: yes, before the first patient, in ClinicalTrials.gov or another WHO registry; systematic reviews: PROSPERO; observational: recommended)?	

B2. Confidentiality

Question	Your answer
Which direct identifiers will you collect (if any), and why?	
How will each patient get a study code?	
Where is the key file (code ↔ identity) kept, and who can open it?	
Where will the de-identified data be stored (institution-approved, password-protected)?	

B3. Feasibility check (tick and comment)

☐	Question	Comment
☐	Time: realistic timeline (ethics 1–3 months, recruitment, analysis, writing)?	
☐	Money: budget headings named (staff time, materials/tests, publication fee)?	
☐	People: who collects data (nights, weekends), who analyses, protected time?	
☐	Patients: how many eligible patients did your unit see last year? How many can you realistically enrol?	
☐	Data: are the key variables recorded routinely? How much is likely missing?	

Biggest feasibility risk: _____ One fix: _____

B4. Publication plan (show it on slide 6 of your presentation)

Item	Your answer
Working title (from your PICO: design, I vs C, P, O; 10–16 words)	
Target journal (and a back-up), checked in DOAJ / NLM Catalog, with fees and licence known	
Reporting guideline (CONSORT / STROBE / STARD / PRISMA / SPIRIT / CARE)	
Registration needed? Where? (from B1)	
Authors (all four ICMJE criteria) and their CRediT roles	
Conflicts of interest and funding to declare	

Item	Your answer
Data-sharing statement (what, where, under which conditions)	
Conference where you would present an abstract first	

Structured-abstract skeleton. One line each. Results are the results you *expect to report* (format, not numbers).

Section	One line
Background	
Methods	
Results (expected)	e.g. "28-day mortality: n/N per group, RR (95% CI)"
Conclusion	Write the sentence you would use if the result is positive, and the one you would use if it is not.

Part C: [L2] Manuscript workshop 1: draft your Methods and Results (50 minutes)

What you are doing

Each group writes the **Methods** and **Results** of a short STROBE-guided original article answering its **FLUID-ICU question**, using **only** its results pack. Nobody calculates anything: every number comes from the pack.

Group	Question (results pack)
1	Q1: Positive fluid balance by day 3 and 28-day mortality (Course/Data/results_packs/Q1/)
2	Q2: Cumulative fluid balance and acute kidney injury (Q2/)
3	Q3: Cumulative fluid balance and ventilator-free days (Q3/)
4	Q4: Initial lactate and 28-day mortality; complete-case vs multiple imputation (Q4/)
5	Q5: Fluid balance and ICU length of stay among survivors (Q5/)

FLUID-ICU is a **simulated teaching dataset**. In your ethics sentence write: *"This analysis used a simulated teaching dataset; no ethics approval or consent was required."*

Roles (agree in the first minute)

Role	Who	Job
Methods writers	2	Design & setting, participants, variables, statistical analysis
Results writers	2	Participants, Table 1 sentence, main result (crude + adjusted), one secondary/sensitivity result

Role	Who	Job
Tables & figures	1	Table 1 + one figure or table from the pack: title (what, in whom, when), n per column, units, footnotes
Number checker	1	Compares every number in the text with the pack; checks STROBE items 4–17

Timing

Minute	Task
0–5	Read your results pack together. Write the main result in ONE sentence (effect size + 95% CI).
5–25	Methods (~300 words), using the headings below.
25–40	Results (~300 words): participants first, numbers with CIs, no interpretation.
40–47	Table 1 + one figure/table with title, legend and footnotes.
47–50	Tick STROBE items; check every number; save and share the file.

Methods: headings and checklist (STROBE items in brackets)

- **Study design and setting** (4, 5): prospective multicentre cohort; 4 ICUs; dates as given in the pack.
- **Participants** (6): who was included and excluded; how many.
- **Variables** (7): exposure (how fluid balance was calculated and over which days), outcome (definition and time point), confounders adjusted for.
- **Data sources** (8): routine ICU charts / study CRF.
- **Bias** (9): how confounding by indication was addressed (adjustment for severity).
- **Study size** (10): "all eligible patients in the cohort were included" (no new sample size for a fixed dataset).
- **Statistical analysis** (12): summary statistics; the test for each comparison; the regression model(s) and which variables; how missing data were handled; software (R, version from the pack); significance level / reporting of 95% CIs.

Results checklist

- ☐ Participants first: number analysed (and excluded, with reasons) (13)
- ☐ Table 1 summarised in one or two sentences: how the groups differed at baseline (14)
- ☐ Outcome numbers per group (15)
- ☐ Main result: **crude AND adjusted** estimate with 95% CI (16)
- ☐ One further analysis (secondary outcome, sensitivity, imputation) (17)
- ☐ No interpretation ("because", "suggests", "important"). Move it to the Discussion
- ☐ No number without its unit and, for effects, its 95% CI

Deliverable

Methods + Results (~600 words) + Table 1 + one figure or table, saved in the shared folder as FLUID-ICU_Q[n]_draft1.